IMO 2026 — Problem 1

Solution by GPT-5.6 Sol (lightweight harness)

2026-07-18

Source: `results/imo-2026/gpt-5.6-sol-lightweight/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

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Approaches tried

- **Lexicographic termination potential (successful):** track the total number of prime factors with multiplicity and, secondarily, the number of nonunit entries. Every move strictly decreases this pair lexicographically.
- **Prime-adic valuation invariant (successful):** for each prime p, a move changes the selected valuations (a, b) into $(\min(a, b), |a - b|)$, preserving the gcd of all p-adic valuations.

- **Product alone (insufficient):** the product of all entries is generally not invariant; it is divided by the selected pair's gcd. It helps motivate the first coordinate of the termination potential but cannot determine the final value.

Current best

Let the initial entries be $a_1, \dots, a_{2026}$. Every play terminates with exactly one nonunit, and its value is

$$\boxed{M = \prod_p p^{\gcd\left(v_p(a_1),\dots,v_p(a_{2026})\right)}},$$

where the product is over all primes and the gcd of an all-zero list is 0. This finite product is determined solely by the initial board.

Full proof

Let the entries currently on the board be $x_1, \dots, x_{2026}$. We first prove termination.

For a positive integer t, let $\Omega(t)$ denote the number of prime factors of t, counted with multiplicity, with $\Omega(1) = 0$. Define

$$S = \sum_{i=1}^{2026} \Omega(x_i), \qquad N = \#\{i : x_i > 1\}.$$

We compare pairs (S, N) lexicographically: the first coordinate is compared first, and the second coordinate is used only when the first coordinates are equal.

Suppose a move selects $m, n > 1$, and put $d = \gcd(m, n)$. The product of the two new entries is

$$d \cdot \frac{\operatorname{lcm}(m,n)}{d} = \operatorname{lcm}(m,n) = \frac{mn}{d}.$$

Since $\Omega(uv) = \Omega(u) + \Omega(v)$ for positive integers u, v, the contribution of these two positions to S changes from $$ (m)+ (n)

to [(m)+ (n)- (d). $$ If $d > 1$, then $\Omega(d) > 0$, so S strictly decreases. If $d = 1$, the new entries are 1 and mn. Thus S stays unchanged, while the two selected nonunits are replaced by exactly one nonunit, so N strictly decreases. Hence every move strictly decreases (S, N) lexicographically.

There cannot be infinitely many such decreases. Indeed, the nonnegative integer S can decrease only finitely many times, and during any interval in which S is fixed, the nonnegative integer N decreases at every move. Therefore every sequence of moves is finite.

When no further move is possible, at most one entry is greater than 1, since any two entries greater than 1 could be selected. On the other hand, a move can never leave no entry greater than 1: if the second new entry

$$\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$$

equals 1, then $\operatorname{lcm}(m,n) = \gcd(m,n)$, which forces $m = n$, and the first new entry is then $m > 1$; if the second new entry is not 1, it is itself greater than 1. Since initially there are entries greater than 1, every configuration reached has at least one such entry. Consequently, every terminal configuration has exactly one entry $M > 1$. This proves part (a).

It remains to show that M is independent of the moves. Fix a prime p. For every positive integer x, write $v_p(x)$ for its exponent in the prime factorization of x. Define

$$G_p = \gcd(v_p(x_1), \dots, v_p(x_{2026})),$$

using the convention that the gcd of an all-zero list is 0.

We claim that G_p is invariant under every move. If

$$a = v_p(m), \qquad b = v_p(n),$$

then the valuations at the two selected positions become

$$v_p(\gcd(m, n)) = \min(a, b)$$

and

$$v_p\left(\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right) = \max(a, b) - \min(a, b) = |a - b|.$$

Assume without loss of generality that $a \le b$. Then

$$\gcd(\min(a, b), |a - b|) = \gcd(a, b - a) = \gcd(a, b).$$

Taking the gcd also with all valuations at the unchanged positions therefore gives the same value before and after the move. Thus G_p is indeed invariant.

Let the initial entries be $a_1, \dots, a_{2026}$. At the end, all entries except M equal 1, so their p-adic valuations are 0. The final value of the invariant is consequently

$$\gcd(v_p(M), 0, \dots, 0) = v_p(M).$$

Its initial value was $\gcd(v_p(a_1), \dots, v_p(a_{2026}))$. Hence, for every prime p,

$$v_p(M) = \gcd(v_p(a_1), \dots, v_p(a_{2026})).$$

By unique prime factorization,

$$M = \prod_p p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

Only primes dividing at least one initial entry can have a nonzero exponent in this product, so it is finite. The formula involves only the initial entries and not the sequence of moves. Therefore the value of M is independent of all choices, proving part (b). □